CO3 - Assignment

P.V.D. Nagendra Reddy
192472399

① Intelligent Agents in a Smart Home

Introduction:

An intelligent agent preceives its environment through sensors and acts on it through actuators to achieve its objectives. The PEAS description of the Smart home.

- Performance : low energy, security, Comfort.
- Environment : rooms, residents, wheather, time.

Types of agents and their function in the System.

→ Simple reflex agent: It uses AND action rules on the current percept only and keeps no history.

- Example: If motion and night then light on. if temperature > 28°c Then Ac on.
- Strength: Very fast and Simple.

→ Model-based reflex agent: It maintains an internal state, a model So it turns world evolves and how its actions affect it.

Example: It remembers that a room has been empty for 2omin, so it turns the light off even when no motion event fires. it remember estimates how fast each room heats up.

- strength: it handles Partial observability.

→ Goal-based agent : It selects action sequences by searching to a goal

• Example : goal "house at 22°C when the user arrives at 6pm". So it works out when to start cooling.

• Weakness : A goal is met or not, so it can't compare two ways of reaching it

→ utility-based agent :- It maximizes a utility function U (state) that scores how desirable a state is, which lets it trade off conflict objectives.

Comparsion

Agent	Memory	Handles trade offs	Learns
mple reflex	No	No	No
del-based	Yes	No	No
l-based	Yes	No	No
-based	Yes	Yes	with a learning element

mended approach : Hybird layered agent with a learning element.

ency reponses with minimal delay.

cy, routiness, thermal behavior.

Preparation, night mode.

the utility weights and the model.

→ Iterative Deepening Search:

- It runs depth-limited DFS with limits 0, 1, 2, ... until the exit is found
- It combines DFS's low memory with BFS's Completeness.
- It is optimal when steps costs are uniform.

Complexity:
(b = branching factor, d = depth of shallowest goal, C^* = optimal cost, ε = minimum step cost)

	Time	Space	Advantages
UCS	$O(b^{(1+LC^*/\varepsilon\rfloor)})$	$O(b^{(1+LC^*/\varepsilon\rfloor)})$	optimal for any positive costs
IDS	$O(b^d)$	$O(b \cdot d)$	Linear memory, Complete optimal for equal costs.

In a maze b is small, so IDS's repeated work is minor, while UCS's memory can run out on a large maze.

Agent design and decision-making.

Agent	Behavior in the maze	Effect.
Simple reflex	Follows a rule such as "turn left"	Loops forever in mazes with cycles, no memory
Model-based	Stores a map of visited cells	Avoids loops and Copes with - Partial observability.
Goal-based	Searches for a Path to the exit	Plans instead of wandering.
Utility-based	Weights Path length, energy and risk	Best when moves have different costs.

The agent's design decides what it remembers and what it optimizes. The search strategy decides how it explores.

V. Best Combination

→ Uniform step costs : A model - based, goal - based agent using IDS. It needs linear memory, which suits a robot's limited hardware, and the stores map prevents revisits.

→ Varying costs : A utility - based agent using UCS, which guarantees the cheapest path at the price of more memory.

Conclusion :-

Pair the agent type with the search strategy. A model - based goal agent with IDS is the practical default, and a utility - based agent with UCS is the choice when cost matters.